Size. Size denotes the greatest point-to-point distance between two motors on a drone. When it comes to a drone, not only weight but size also varies from small to large based on requirements.

Cost of the drone always depends on its functionalities and application, regardless of size. Small sizes, say, below 600mm, are good to start with if you are still learning.

Selection of components

Choosing appropriate components is crucial to building a drone with desired specifications. A few tips for different components are mentioned in this section.

Frame. A frame is the basic skeleton of a drone for attaching all other components as well as providing protection. Commonly-known configurations include quadcopter (four arms), tri-copter (three arms), hexa-copter (six arms), octocopter (eight arms) and so on. Each arm is connected to a motor.

Besides configuration, material chosen decides durability of the drone. Such materials as plastic, carbon-fibre, PCB and aluminium are generally used. Though expensive, carbon-fibre is the best option as it is durable and lightweight.

- As a beginner, prefer relatively simpler quadcopter (x or +) or tri-copter frames, such as in DJI Phantom 2 Vision or Cheerson CX-33S. These frames can be rebuilt easily after crashing. Most flight controllers can work with quadcopter frames. Tri-copter frames, although cheap, need more work than quadcopters due to lack of symmetry.
- For commercial applications, a hexa-copter or octocopter is the way to go. These have more motors and larger batteries, and are ideal for lifting heavy loads. This implies that even if a motor or propeller fails, the drone can land safely.
- For smaller drones, you can utilise a 3D printer to get the right frame shape when working with plastic. PCB also works well for small sizes.
- While mounting the components, keep in mind that carbon-fibre hampers radio frequency (RF) signals.

Motors. Motors generate force to rotate the propellers and move through the air. The more efficient the motor, the longer the battery life and, hence, the flight.

- All motors should have the same thrust rating, usually given in grams, for ensuring stability.
- Some motors have performance comparison charts that can be utilised to make sure that the necessary target is met, especially in commercial drones.
- In large drones, brushless out-runner type motors (like FPVDrone 1104 motor) are used, while in smaller, inexpensive models, brushed motors are preferred. Unlike brushed motors whose brushes wear out quickly, brushless motors have longer lifespans, better durability and the ability to generate more power at low noise.
- Always check for kV rating in specifications. If you want the motor to spin faster, choose a motor with higher kV rating. Usually, a larger motor with lower kV rating has better efficiency and stability than a smaller one with high kV rating.